Advance Mobile Application Development

File management in flutter

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Working with Files in Flutter

- By using the path_provider plugin we can get the path to common directories designed for different purposes:

```
import 'package:path_provider/path_provider.dart';


// Put cache files in this directory
final temporaryDirectory = await getTemporaryDirectory();

// For files that our app uses but are not exposed to the user
final appSupport = await getApplicationSupportDirectory();

 // For user-generated files
final appDocuments = await getApplicationDocumentsDirectory();
```

Create a file

- Let's create a random file in the temporary directory and write something inside:

```
final Directory tempDir = await getTemporaryDirectory();

final File file = File('${tempDir.path}/sample_file.txt');


await file.writeAsString('Sample content to write');
```

Delete a file

- The delete method is just as crucial as the create method, as it helps clear out old files and thus saves valuable storage space.

```
final Directory tempDir = await getTemporaryDirectory();

final File file = File('${tempDir.path}/sample_file.txt');


await file.delete();
```

Create a directory

- Now let's say that I want to create a new file, but within a certain directory:

```
final Directory tempDir = await getTemporaryDirectory();
final Directory newDirectory = Directory('${tempDir.path}/sample_directory');

// Always check that the directory exists
if (await newDirectory.exists() == false) {
await newDirectory.create();
}

final File file = File('${newDirectory.path}/sample_file.txt');

await file.writeAsString('Sample content to write');
```

Remove a directory

- Now let's do the reverse and delete the directory:

```
final Directory tempDir = await getTemporaryDirectory();

final Directory newDirectory = Directory('${tempDir.path}/sample_directory');


await newDirectory.delete(recursive: true);
```

Read file

- Let's open a file to see its content:

```
final Directory tempDir = await getTemporaryDirectory();

final File file = File('${tempDir.path}/sample_file.txt');


final String fileContent = await file.readAsString();
```

Copy file

- Now, let's generate a duplicate of the previously created sample file:

```
final Directory tempDir = await getTemporaryDirectory();

final File file = File('${tempDir.path}/sample_file.txt');


 final File copy = await file.copy('${tempDir.path}/copy_file.txt');
```

Rename file

- Next, let's change the name of the file we just copied:

```
final Directory tempDir = await getTemporaryDirectory();

final File file = File('${tempDir.path}/copy_file.txt');


await file.rename('${tempDir.path}/new_name.txt');
```

Permission to write

- To allow the user to read and write from the External Storage we should add the permissions in AndroidManifest.xml
- android/app/src/main/AndroidManifest.xml

```
<uses-permission android:name="android.permission.WRITE_EXTERNAL_STORAGE"/>
<uses-permission android:name="android.permission.READ_EXTERNAL_STORAGE"/>
```

```
<application
    android:requestLegacyExternalStorage="true"
```

Cont.

- We should also make sure the permission to write is Granted before we start writing a file to the External storage.

```
var status = await Permission.storage.status;
if (!status.isGranted) {
  print("no permistiom");
  await Permission.storage.request();
}

// now we can write our files
await file.writeAsString(containt);
```

An example

- As an example we have created a dart file called **textfiles.dart**
- We a class called Textfile{}
- Containing few functions such as the shown here ->

ADME.md | main.dart | textfiles.dart | AndroidManifest.xml | pubspec.yaml

```dart
import 'dart:convert';
import 'dart:io';
import 'main.dart';
import 'package:permission_handler/permission_handler.dart';
import 'package:path_provider/path_provider.dart';

class Textfiles {

  readTextFile() async{...}

  Future<String> readTextFile2(File? filepath) async{...}

 saveTextFile(String filename, String containt) async{...}

 creatDir(String dirname) async {...}void

 getthepath() async {...}

}
```

```
readTextFile() async{
  final Directory tempDir = await getTemporaryDirectory();
  final File filepath = File('${tempDir.path}/file_picker/a text file 005.txt');
  final String fileContent = await file.readAsString();
  return(fileContent);
}
```

- **getTemporaryDirectory();** we find the Temporary Directory and store it in the variable **tempDir** from type Directory
- **File** will compaine the Directory along with the file name and store it in variable filepath from type File
- **readAsString();** will read the file given in the variable filepath and store the content in a String variable called fileContent which is the return value from readTextFile() function
-

```
Future<String> readTextFile2(File? filepath) async{

final String fileContent = await filepath!.readAsString();
 return(fileContent);
   }
```

- **readTextFile2(File? Filepath)**
- Reads a specific text file which we passed the file path to.

```
Future<String> readTextFile3(File? filepath) async{

final List<String> fileContent2 = await filepath!.readAsLines();

 int count=0;
 for (var i in fileContent2){
   //print(i);
   if ((count==4) || (count > 6)){
     List<String> s=i.split(",");
     print(s[1]);
   }
```

- **readTextFile2(File? Filepath)**
- Reads the comma separated file line by line and prints the second value in each line

- **saveTextFile(String filename, String containt)**
- **writeAsString** writes the String Variable content in the file given in the variable **filepath** which we pass the both to the function **saveTextFile** as parameters.

```
saveTextFile(String filename, String content) async{

final File filepath = File('/storage/emulated/0/Documents/'+filename+".text");

  var status = await Permission.storage.status;
  if (!status.isGranted) {
    print("no permistiom");
    await Permission.storage.request();
  }

  await filepath.writeAsString(content);
}
```

- creatDir(String dirname) creates a new Directory with a test text file.
- We should check if the directory already exists then we don't need to create it again

```
creatDir(String dirname) async {
    ///////////here we creat a file in side our own Dir in temp folder
final Directory newDirectory = Directory('/storage/emulated/0/'+dirname);

// Always check that the directory exists
    if (await newDirectory.exists() == false) {
      await newDirectory.create();
    }

    final File file = File('${newDirectory.path}/sample_file.txt');
    print(""+tempDir.toString());
    await file.writeAsString('this file writen as the try ');
  }
```

- **FilePicker:** we use FilePicker to pick a file from the storage and read it where FilePicker.*platform*.pickFiles(); will allow us to browses the device storage and select a file and store it in temporary directore associated with our app.

```
import 'package:file_picker/file_picker.dart';
```

```
ElevatedButton(
  onPressed: () async {
    try {
      result = await FilePicker.platform.pickFiles();
      if (result != null) {
      if (!kIsWeb) {
        file = File(result!.files.single.path!);
        filecontaint = (await fileobj.readTextFile2(file)) as Future<String>;
        }
      } else {    print("user cancled the picker"); } // User canceled the picker
    } catch (_) {}
    super.setState(()  { });
    },
  child: const Text('Pick a Text file File'), ),
```